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Reply-to: Guy Davidson

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Many thanks to the following people who have helped to bring this proposal to life:

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Reproducible floating-point results

1. Abstract

The C++ standard, hereafter referred to as [C++], has very little to say about floating-point specification, which means that it is extremely hard to write floating-point expressions that, given the same inputs and rounding mode, will produce the same results on all implementations. This paper proposes the introduction of a new type which specifies sufficient conformance with the provisions specified in ISO/IEC 60559:2020 (which is itself imported from IEEE Std 754-2019, unmodified except for editorial adjustment), hereafter referred to as [559], to guarantee this reproducibility.

2. Revision history

R0 2024-09-09

Presented to the BSI C++ panel on 2024-09-30

R1 2024-10-08

- Extended the contributor list. Many thanks, all of you.
- Clarification of the relationship between IEE-754:2019 and ISO/IEC 60559:2020.
- Added revision list.
- Expanded motivation, clarifying non-associativity and varying error requirements.
- Added banking use case.
- Clarification of the need for bitwise identical results in simulations for games.
- Added geometry use case.
- Highlighted para 20 of Annex F.3 of the C standard.
- Outlined non-goals.
- Additional section on the provision of correctly rounded mathematical functions.